Recitation 2

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Exercise 1

(a) Fix $L \in \mathbb{R}$.

Assume A holds. Then there exists $\epsilon_0 > 0$ and $N \in \mathbb{N}$ such that

$$|z_n - L| \geq \epsilon_0$$

for every $n \geq N$. Suppose, for contradiction, that $z_n \to L$. Then there exists $N_0 \in \mathbb{N}$ such that

$$|z_n - L| < \epsilon_0$$

for every $n \geq N_0$. If we choose $n \geq max\{N, N_0\}$, then we have

$$|z_n - L| \geq \epsilon_0 \qquad \text{and} \qquad |z_n - L| < \epsilon_0$$

at the same time, which is a contradiction. Therefore A implies divergence of $\{z_n\}$

Assume B holds. Then there exists $\epsilon_0 > 0$ such that, for every $N \in \mathbb{N}$, there exists $n \geq N$ with

$$|z_n - L| \geq \epsilon_0.$$

Suppose, for contradiction, that $z_n \to L$. Then there exists $N_0 \in \mathbb{N}$ such that

$$|z_n - L| < \epsilon_0$$

If we choose $n \geq max\{N, N_0\}$, then we have

$$|z_n - L| \geq \epsilon_0 \qquad \text{and} \qquad |z_n - L| < \epsilon_0$$

at the same time, which is a contradiction. Therefore B implies divergence of $\{z_n\}$

For C, let

$$z_n = L + \frac{1}{n}.$$

Then

$$|z_n - L| = \frac{1}{n}.$$

Let $\epsilon > 0$ and choose $N \in \mathbb{N}$ such that

$$N > \frac{1}{\epsilon}.$$

For every $n \geq N$,

$$|z_n - L| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

Therefore, $z_n \to L$. Given any $N \in \mathbb{N}$, choose $n = N$, and define

$$\varepsilon_n = \frac{1}{2n} > 0.$$

Then

$$|z_n - L| = \frac{1}{n} \geq \frac{1}{2n} = \varepsilon_n.$$

Thus C is true even though $\{z_n\}$ converges to L. Therefore, C does not imply that $\{z_n\}$ diverges.

Recall that $\{z_n\}$ converges to L if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N, \qquad |z_n - L| < \varepsilon.$$

Negating this statement gives

$$\exists \varepsilon_0 > 0,\ \forall N \in \mathbb{N},\ \exists n \geq N, \qquad |z_n - L| \geq \varepsilon_0.$$

Thus, statement B is the exact negation of convergence to L.

No. A sequence can fail to converge to one particular L while converging to a different real number. For example, let

$$z_n = 1 \qquad \text{for every } n \in \mathbb{N}.$$

Then $\{z_n\}$ converges to 1. But it does not converge to 0, since

$$|z_n - 0| = 1$$

for every n. Thus, failure to converge to one particular value does not imply divergence.

(b) The definition of a divergent sequence is

$$\exists \varepsilon_0 > 0,\ \forall N \in \mathbb{N},\ \exists n \geq N, \qquad |z_n - L| \geq \varepsilon_0.$$

Consider the following sequence

$$a_n = 1 + (-1)^n.$$

Notice that this simplifies to

$$a_n = \begin{cases} 2, & n \text{ even}, \\ 0, & n \text{ odd}. \end{cases}$$

Let $L \in \mathbb{R}$. Set

$$\varepsilon_0 = 1.$$

Let $N \in \mathbb{N}$. We consider two cases.

If $L < 1$, choose an even integer $n \geq N$. Then $a_n = 2$, so

$$|a_n - L| = |2 - L| = 2 - L > 1 = \varepsilon_0.$$

If $L \geq 1$, choose an odd integer $n \geq N$. Then $a_n = 0$, so

$$|a_n - L| = |0 - L| = |L| \geq 1 = \varepsilon_0.$$

Thus, for every $L \in \mathbb{R}$, we have found $\varepsilon_0 = 1$ such that for every $N \in \mathbb{N}$, there exists $n \geq N$ with

$$|a_n - L| \geq \varepsilon_0.$$

Therefore, for every $L \in \mathbb{R}$, the sequence does not converge to L.

Exercise 2 Let $\{x_n\}_{n\in\mathbb{N}}$ be a real sequence such that $x_n^2 \to 4$.

(a) Consider $||x_n| - 2|$. Then

$$\begin{aligned} ||x_n| - 2| &= \frac{|x_n^2 - 4|}{|x_n| + 2} \\ &\leq \frac{|x_n^2 - 4|}{2} \end{aligned}$$

Let $\epsilon > 0$. Since $x_n^2 \to 4$, there exists $N \in \mathbb{N}$ such that for $n > N$, we have

$$|x_n^2 - 4| < 2\epsilon.$$

Hence, for $n \geq N$,

$$||x_n| - 2| \leq \frac{|x_n^2 - 4|}{2} < \epsilon.$$

Therefore,

$$|x_n| \to 2.$$

(b) No, consider $\{x_n\} = 2(-1)^n$. Then

$$x_n^2 = 2^2(-1)^{2n} = 4$$

for every $n \in \mathbb{N}$. So $x_n^2 \to 4$. However, this sequence simplifies to

$$x_n = \begin{cases} 2, & n \text{ even}, \\ -2, & n \text{ odd}. \end{cases}$$

That is, it oscillates between -2 and 2, and so it never converges.

(c) Suppose that there exists $n_0 \in \mathbb{N}$ such that $x_n x_{n+1} \geq 0$ for every $n \geq n_0$.

Choose $\epsilon = 1$. Since $|x_n| \to 2$, there exists $N_1 \in \mathbb{N}$ such that for every $n \geq N_1$,

$$||x_n| - 2| < 1.$$

Furthermore, $|x_n| > 1 \neq 0$.

Let $N = \max\{N_1, n_0\}$. Then $x_n \neq 0$, $x_{n+1} \neq 0$, and

$$x_n x_{n+1} > 0$$

for every $n \geq N$. Thus x_n and x_{n+1} have the same sign.
If $x_n > 0$ for all $n \geq N$, then $x_n = |x_n|$. So

$$x_n \to 2.$$

If $x_n < 0$ for all $n \geq N$, then $x_n = -|x_n|$. So

$$x_n \to -2.$$

Therefore $\{x_n\}$ either converges to 2 or -2.

Exercise 3 Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a bijection, and define $y_n = x_{\sigma(n)}$ for a real sequence $\{x_n\}_{n\in\mathbb{N}}$.

(a) Suppose $x_n \to L$. Let $\epsilon > 0$. Then there exists $N \in \mathbb{N}$ such that

$$|x_m - L| < \epsilon$$

for all $m \geq N$. Since σ is a bijection, the set

$$\{n \in \mathbb{N} : \sigma(n) < N\}$$

is finite. So there exists $M \in \mathbb{N}$ such that for all $n \geq M$, we have

$$\sigma(n) \geq N.$$

Therefore, for all $n \geq M$, we have

$$|y_n - L| = |x_{\sigma(n)} - L| < \epsilon.$$

That is

$$y_n \to L.$$

Conversely, suppose $y_n \to L$. Let $\epsilon > 0$. Then there exists $N \in \mathbb{N}$ such that

$$|y_m - L| < \epsilon$$

for all $m \geq N$. Since σ is a bijection, there exists a bijective inverse σ^{-1}. Moreover,

$$y_{\sigma^{-1}(n)} = x_n.$$

Because σ^{-1} is also a bijection, the set

$$\{n \in \mathbb{N} : \sigma^{-1}(n) < N\}$$

is finite. So there exists $M \in \mathbb{N}$ such that for all $n \geq M$, we have

$$\sigma^{-1}(n) \geq N.$$

Therefore, for all $n \geq M$, we have

$$|x_n - L| = |y_{\sigma^{-1}(n)} - L| < \epsilon.$$

That is

$$x_n \to L.$$

(b) No, let

$$x_1 = 1 \qquad \text{and} \qquad x_n = 0, \text{ for all } n \geq 2.$$

So

$$x_n \to 0.$$

Define $\sigma : \mathbb{N} \to \mathbb{N}$ by

$$\sigma(n) = \begin{cases} 1, & n \text{ odd}, \\ n/2, & n \text{ even}. \end{cases}$$

Then σ is surjective since

$$\sigma(2k) = k$$

for all $k \in \mathbb{N}$. But σ is not injective since

$$\sigma(1) = \sigma(2) = 1.$$

Let

$$y_n = x_{\sigma(n)}.$$

Since $\sigma(n) = 1$ for all odd n,

$$y_n = 1, \text{for all odd } n \implies y_n \not\to 0.$$

But $x_n \to 0$. Therefore, σ must be bijective in order for $x_n \to L \implies y_n \to L$ to hold.

Exercise 4

Let

$$a_n = \frac{(-1)^{n+1}}{n}, \qquad s_n = \sum_{j=1}^{n} a_j.$$

Assume $s_n \to S$ for some $S > 0$. Define

$$b_{3k-2} = \frac{1}{4k-3}, \qquad b_{3k-1} = \frac{1}{4k-1}, \qquad b_{3k} = -\frac{1}{2k} \qquad \forall k \in \mathbb{N}$$

and let $t_n = \sum_{j=1}^{n} b_j$.

(a) Define $\sigma : \mathbb{N} \to \mathbb{N}$ by

$$\sigma(3k-2) = 4k-3, \qquad \sigma(3k-1) = 4k-1, \qquad \sigma(3k) = 2k.$$

for all $k \in \mathbb{N}$. Then

$$a_{\sigma(3k-2)} = a_{4k-3} = \frac{1}{4k-3} = b_{3k-2},$$

$$a_{\sigma(3k-1)} = a_{4k-1} = \frac{1}{4k-1} = b_{3k-1},$$

and

$$a_{\sigma(3k)} = a_{2k} = -\frac{1}{2k} = b_{3k}.$$

Notice that the function outputs,

$$\sigma(3k-2) = 4k-3 \equiv 1 \pmod 4,$$

$$\sigma(3k-1) = 4k-1 \equiv 3 \pmod 4,$$

and

$$\sigma(3k) = 2 = 2k \equiv 0 \text{ or } 2 \pmod 4,$$

are disjoint and cover all of $\mathbb{N}$. Thus σ is a bijection.

(b) Notice that t_{3m} can be broken down into

$$\begin{aligned} t_{3m} &= \sum_{k=1}^{m} \left(\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} \right) \\ &= \sum_{k=1}^{2m} \frac{1}{2k-1} - \sum_{k=1}^{m} \frac{1}{2k}. \end{aligned}$$

Similarly for s_{4m}, we have

$$\begin{aligned} s_{4m} &= \sum_{k=1}^{2m} \frac{1}{2k-1} - \sum_{k=1}^{2m} \frac{1}{2k} \\ &= \sum_{k=1}^{2m} \frac{1}{2k-1} - \sum_{k=1}^{m} \frac{1}{2k} - \sum_{k=m+1}^{2m} \frac{1}{2k}. \end{aligned}$$

Now for s_{2m}, we have

$$\begin{aligned} s_{2m} &= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{1}{2m-1} - \frac{1}{2m} \\ &= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{2m} \right) - 2 \left(\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2m} \right) \\ &= \sum_{j=1}^{2m} \frac{1}{j} - 2 \sum_{k=1}^{m} \frac{1}{2k} \\ &= \sum_{j=1}^{2m} \frac{1}{j} - \sum_{k=1}^{m} \frac{1}{k} \\ &= \sum_{k=m+1}^{2m} \frac{1}{k}. \end{aligned}$$

Thus

$$\frac{1}{2} s_{2m} = \sum_{k=m+1}^{2m} \frac{1}{2k}$$

Hence

$$\begin{aligned} s_{4m} + \frac{1}{2} s_{2m} &= \left(\sum_{k=1}^{2m} \frac{1}{2k-1} - \sum_{k=1}^{m} \frac{1}{2k} - \sum_{k=m+1}^{2m} \frac{1}{2k} \right) + \sum_{k=m+1}^{2m} \frac{1}{2k} \\ &= \sum_{k=1}^{2m} \frac{1}{2k-1} - \sum_{k=1}^{m} \frac{1}{2k} \\ &= t_{3m} \end{aligned}$$

(c) Since $s_n \to S$, every subsequence of $\{s_n\}$ also converges to S. Therefore,

$$s_{4m} \to S \qquad \text{and} \qquad s_{2m} \to S.$$

By part (b),

$$t_{3m} = s_{4m} + \frac{1}{2}s_{2m}.$$

Thus, by the algebraic limit theorem,

$$t_{3m} \to S + \frac{1}{2}S = \frac{3S}{2}.$$

Let $\epsilon > 0$. Since

$$t_{3m} \to \frac{3S}{2},$$

there exists $M_1 \in \mathbb{N}$ such that, for every $m \geq M_1$,

$$\left| t_{3m} - \frac{3S}{2} \right| < \frac{\epsilon}{2}.$$

Also, choose $M_2 \in \mathbb{N}$ such that

$$M_2 > \frac{1}{\epsilon}.$$

Then, whenever $m \geq M_2$,

$$\begin{aligned} \frac{1}{4m+1} + \frac{1}{4m+3} &< \frac{2}{4m+1} \\ &< \frac{1}{2m} \\ &< \frac{\epsilon}{2}. \end{aligned}$$

Let

$$M = \max\{M_1, M_2\}.$$

Now let $n \geq 3M$. Then n has one of the forms

$$n = 3m, \qquad n = 3m+1, \qquad \text{or} \qquad n = 3m+2,$$

where $m \geq M$. If $n = 3m$, then

$$\left| t_n - \frac{3S}{2} \right| = \left| t_{3m} - \frac{3S}{2} \right| < \frac{\epsilon}{2} < \epsilon.$$

If $n = 3m+1$, then

$$t_{3m+1} = t_{3m} + \frac{1}{4m+1}.$$

Therefore,

$$\begin{aligned} \left| t_{3m+1} - \frac{3S}{2} \right| &\leq \left| t_{3m} - \frac{3S}{2} \right| + \frac{1}{4m+1} \\ &< \frac{\epsilon}{2} + \left(\frac{1}{4m+1} + \frac{1}{4m+3} \right) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon. \end{aligned}$$

If $n = 3m + 2$, then

$$t_{3m+2} = t_{3m} + \frac{1}{4m+1} + \frac{1}{4m+3}.$$

Hence,

$$\begin{aligned}\left|t_{3m+2} - \frac{3S}{2}\right| &\leq \left|t_{3m} - \frac{3S}{2}\right| + \frac{1}{4m+1} + \frac{1}{4m+3} \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} \\ &= \epsilon.\end{aligned}$$

Thus, for every $n \geq 3M$,

$$\left|t_n - \frac{3S}{2}\right| < \epsilon.$$

Therefore,

$$t_n \to \frac{3S}{2}.$$

(d) Question 3 concerns the individual terms of the sequences, while question 4 concerns their partial sums. The sequence b_n is a permutation of a_n, so both sequences of terms converge to 0. However, s_n and t_n are the partial sums of these terms, not permutations of each other. Therefore, Question 3 does not require s_n and t_n to have the same limit, so there is no contradiction.